Phase 2 - Proposal and Code Implementation - Gopika Devabalaji

Project Title

Explainable Plant Disease Detection System Using CNN and Transfer Learning

Background and Motivation

Plant diseases are among the main causes of economic losses and reduced agricultural productivity worldwide. Detection of plant diseases at early stage can help the farmers to take preventive measures and increase the yield of crops. With the progress in Artificial Intelligence and Deep Learning, Convolutional Neural Networks (CNNs) have shown promising performance in image classification tasks. The main aim of this project is to build an automated plant disease detection system using CNN based models and to help in detection of diseases from plant leaf images.

Problem Statement

Manual detection of plant diseases is time-consuming and requires expert knowledge. The purpose of this project is to develop an image classification system using CNN that can automatically identify plant disease from the image of the leaves. The performance of a custom CNN model will be compared with an architecture based on MobileNetV2.

Dataset link and selected dataset

Dataset Name: PlantVillage

Dataset Link:

https://www.kaggle.com/datasets/emmarex/plantdisease

Description of Dataset

The PlantVillage dataset is a collection of images of healthy and unhealthy plant leaves in different crop categories. This dataset is used frequently in the research of plant disease classification.

Dataset Statistics

• Total images: 20,639

•15 Classes

• Image Type: RGB Color Image

• Types: Bell Pepper, Potato and Tomato plant leaves

• Type of Classification: Multi-class Image Classification

Class Distribution

The dataset has samples of leaves that're healthy and leaves that are diseased. The classes are things, like spot, early blight, late blight, leaf mold, septoria leaf spot, spider mites, target spot, yellow leaf curl virus, mosaic virus and healthy leaves.

Research Questions

1. How well can CNN models identify plant diseases from leaf photos?

2. Is a custom CNN model better or worse than using MobileNetV2?

3. Which plant diseases are identified correctly?

4. What are the common mistakes made when identifying plant diseases?

5. How can we make AI models, for plant diseases more understandable?

Expected Results

• Effective plant disease classification using deep learning.

• Custom CNN and MobileNetV2 models that achieve good classification performance.

• Creation of confusion matrix, classification report , performance curves.

• Identified strengths and limitations of each model.

• Basis for future explainability analysis using Grad-CAM.

Proposed Methodology

Step 1: Collecting Dataset

We will use the PlantVillage dataset from Kaggle for training and evaluation. The PlantVillage dataset is a resource.

Step 2: Preparing Data

To prepare our data we will do the following:

• Resize the images to a size

• Normalize the pixel values to a common range

• Split the dataset into training and validation sets

Step 3: Enhancing Data

To help our model generalize better we will try the following:

• Rotate the images a bit

• Zoom in and out of the images

• Flip the images horizontally

• Shift the images, in width and height

Step 4: Model Development

We will create a Custom CNN Model. This Custom CNN Model will have parts, including:

* Convolution Layers to extract features

* Max Pooling Layers to reduce dimensions

* Dropout Layer to prevent overfitting

* Dense Layers to make predictions

* Softmax Output Layer to give us the probabilities. The Custom CNN Model will help us classify images.

The model will use the PlantVillage dataset to learn and make predictions.We will also use a Transfer Learning Model.

This Transfer Learning Model uses the MobileNetV2 Architecture.

The MobileNetV2 Architecture is good for a feature extraction approach.

We will add a classification head to the Transfer Learning Model for disease prediction.

Step 5: Model Training

We will train the Custom CNN Model and the Transfer Learning Model using things.

These things are

* Adam Optimizer

* Categorical Crossentropy Loss

* Accuracy Metric.

Step 6: Model Evaluation

We will see how well the Custom CNN Model and the Transfer Learning Model work.

We will do this by looking at

* Accuracy

* Precision

* Recall

* F1-Score

* Confusion Matrix

* Classification Report.

This will help us understand how good the Custom CNN Model and the Transfer Learning Model really are, at disease prediction.

Evaluation Metrics

• Accuracy

• Precision

• Recall

• F1-Score

- Confusion Matrix
- Classification Report

Expected Figures

1. I want to see a chart that shows how classes are distributed.
2. I need some sample images of plant leaves.
3. A curve showing how accurate the CNN model is.
4. A curve showing the loss of the CNN model.
5. I also want to see a curve showing how accurate the MobileNetV2 model is.
6. A curve showing the loss of the MobileNetV2 model.
7. Confusion Matrix
8. A chart that compares the performance of models.

Expected Tables

1. A summary of the dataset, in a table.
2. A table that summarizes the architecture of the model.
3. A table that compares the performance of models.
4. A table that lists the evaluation metrics.

Conclusion

The project is about an automatic plant disease detection system using CNN and transfer learning techniques. The developed models will be evaluated using standard metrics of machine learning and the results will provide insight into the effectiveness of deep learning for classification of agricultural diseases.